Employment Continuity Around Childbirth in Japan's Dual Labour Market

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September 4, 2026

Japan: generous leave, large interruptions, a divided labour market

Japan combines three features that are hard to reconcile:

- **Comparatively generous formal leave entitlements**: childcare leave can extend until the child turns two when, for example, nursery admission is unavailable (MHLW; OECD PF2.1).
- **High employment interruption around childbirth**: maternal employment is considerably lower when the youngest child is aged 0–2 than when children are older (OECD Family Database, LMF1.2).
- **A strongly segmented labour market**: a durable divide between regular (*seishain*) and non-regular (*hi-seiki*) employment, particularly important for women (OECD, 2024).

When women stop working around childbirth, do they remain attached to their jobs?

Two margins: working now, and keeping the job

I separate two outcomes that can be combined into one "employment" measure:

A. **Current work** - is the woman doing paid work in the survey reference month?

B. **Employment attachment** - is she working *or* temporarily absent while retaining a job?

Question. Does pre-birth contract status predict the distinction between a *temporary work interruption* and a *loss of employment attachment*?

Preview. Current work is much lower in the birth-report wave for both regular and non-regular women. The regular/non-regular divide appears in whether that interruption occurs inside an ongoing employment relationship or through loss of attachment.

Related literature: Kleven, Landais and Søgaard (2019); Kleven et al. (2019); Goldin (2014); Yamaguchi (2019); Kikuchi (2026).

Japan's dual labour market I

Regular employment (seishain)

- typically open-ended, career-track employment;
- stronger employment continuity;
- greater practical capacity to maintain an employer relationship through an absence.

Non-regular employment (hi-seiki)

- includes part-time, fixed-term, dispatch and other non-regular positions;
- weaker employment continuity;
- formal leave rights may not translate into the same effective job continuity.

Institutional background: OECD, 2024; JILPT, 2024.

2024: 14.4 million women in non-regular employment and 13.0 million in regular employment. Women make up roughly two-thirds of Japan's non-regular workforce (MHLW, Labour Economy White Paper 2025).

Regular/non-regular is not the same as full-time/part-time. Official employment type is based on the type of position at the workplace, not simply on hours worked (Statistics Bureau of Japan, 2022).

Data and measurement

Data. Harmonised KHPS/JHPS household panels, 2004–2022.

Sample. 631 women observed around a birth-report transition, after excluding 31 independently confirmed higher-parity birth reports.

Event time. $t = 0$ is the annual survey wave in which the household reports the focal birth.

$$t = -2 \;\rightarrow\; t = -1 \;\rightarrow\; \underbrace{t = 0}_{\text{birth-report wave}} \;\rightarrow\; t = +1 \;\rightarrow\; \cdots$$

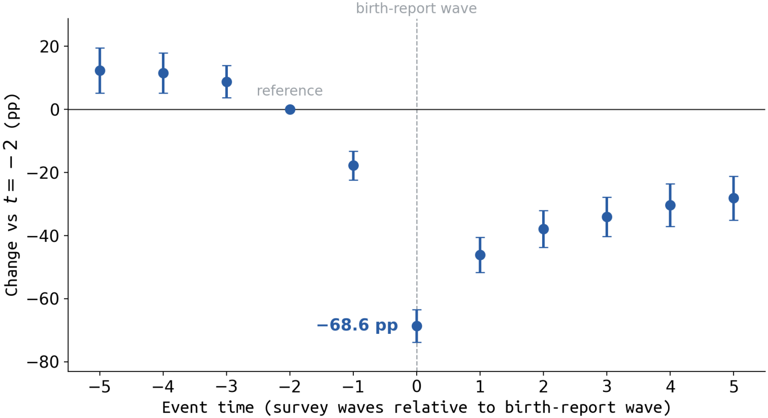
Event time is therefore measured in survey waves rather than from an exact delivery date.

Outcomes.

- **Current work**: reported paid work in the survey reference month.
- **Employment attachment**: current work or temporary absence while retaining a job.

A woman on childcare leave can therefore be recorded as not currently working while remaining employment-attached.

Current work around the birth-report wave

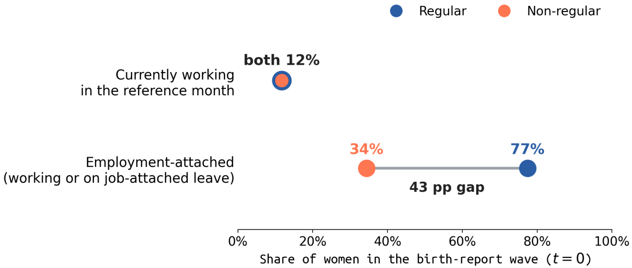


Current work is lowest in the birth-report wave.

Coefficients relative to $t = -2$, with calendar-year fixed effects and episode-clustered 95% CIs.

The panel is unbalanced; $t = -2$ is a presentation reference rather than an untreated baseline.

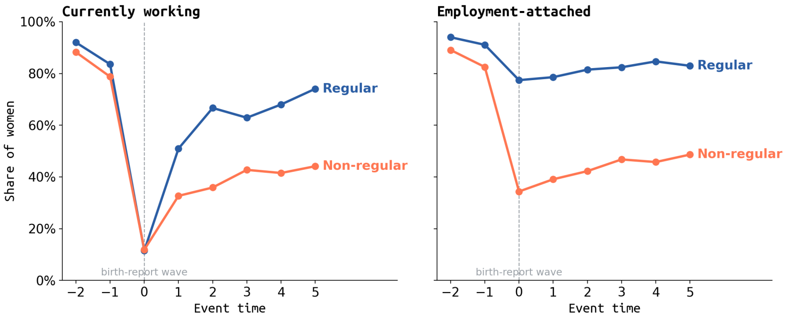
Same interruption, different employment continuity



Among attached non-workers at $t = 0$, **194 of 205 (94.6%) report childcare leave** as the reason for their absence.

Raw shares in the birth-report wave by pre-birth contract group. The group is the last valid wage-job classification observed in $t = -5, \ldots, -1$ and is then held fixed. The later risk-set models use contract measured specifically at $t = -1$. Annual maternity leave is not separately identified.

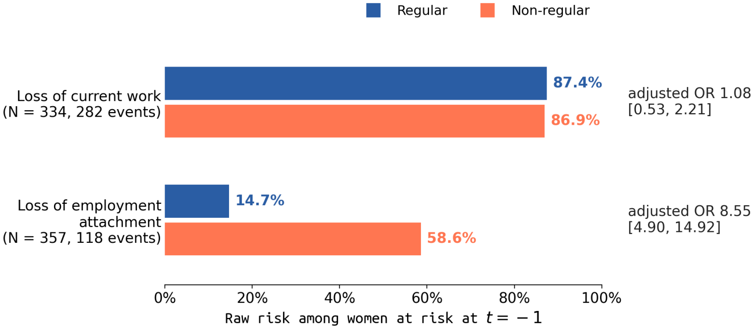
Regular workers retain attachment; non-regular workers often do not



Raw shares over event time by fixed pre-birth contract group (last valid wage-job classification in $t = -5, \ldots, -1$). Current work is lowest at $t = 0$ in both groups, with faster recovery among regular workers. Employment attachment remains around 77–85% for regular workers after $t = 0$; for non-regular workers it falls to 34% and reaches only 49% by $t = +5$. Long-horizon shares are composition-sensitive in the unbalanced panel.

Contract status predicts attachment loss

Risk sets: women at risk at $t = -1$ and observed at $t = 0$.



Adjusted ORs from missing-indicator logits controlling for small firm size. Model-standardised attachment-loss probabilities are 15.3% for regular workers and 58.6% for non-regular workers.

Employment relationships diverge after birth

	Pre-birth contract	$t = +1$	$t = +3$	$t = +5$
Currently working	Regular	51.0%	62.9%	74.1%
	Non-regular	32.7%	42.7%	44.1%
Still has a job	Regular	78.6%	82.4%	83.0%
	Non-regular	39.1%	46.8%	48.6%

- Regular workers return to current work while retaining a high level of job continuity: around 79–83% still have a job at each post-birth horizon.
- Non-regular workers are much less likely to retain a job: 34% do so in the birth-report wave, rising to about 49% by $t = +5$.

"Still has a job" includes women who are currently working and women temporarily absent while retaining an employment relationship. Contract group is the last valid wage-job classification in $t = -5, \ldots, -1$ and is held fixed thereafter. Raw shares are descriptive and composition-sensitive in the unbalanced panel.

Conclusion: leave or exit

1. **Current work is very low in the focal birth-report wave in both contract groups:** around 12% of regular and non-regular women are working at $t = 0$.
2. **Employment continuity differs substantially by contract type:** 77% of regular workers still have a job, compared with 34% of non-regular workers.
3. **Among women attached in the preceding wave,** non-regular status is associated with about a 43 percentage-point higher probability of losing that employment relationship (adjusted OR ≈ 8.6).

Employment continuity through childbirth is highly unequal across Japan's dual labour market.

Among women who retain a job while absent from work, **94.6% report childcare leave**.

Thank you

Thank you.

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A. Sample construction

1,697
complete candidate birth-report records

↓

1,194
first-observed candidate events

↓

662
legacy interval-eligible cohort

−31 independently confirmed higher-parity events

↓

631 women
corrected event cohort

4,923	**4,480**	**627**
person-waves	determinate observations	focal-woman episodes

B. Outcome definitions and questionnaire routing

- **Current work**: questionnaire-reported paid work in the reference month; contradictory responses remain missing. Temporary absence while retaining a job does *not* count as currently working.
- **Employment attachment**: current work or verified job-attached temporary absence. Childcare leave is one identified reason for job-attached absence; annual maternity leave is not separately identified, so attached non-workers are labelled "employment-attached", not "on leave".
- **Weekly hours**: the job block is asked of women who worked and women temporarily absent while retaining a job. For the latter, reported hours describe the job's usual hours, not realised hours in the reference month. Hours are never used to define employment.

Routing diagnostic. Of 4,923 person-waves: 49.9% observed positive hours, 38.9% structural skips, 3.8% refusal/don't-know, 7.0% focal-woman role unavailable, 0.5% contradictory. All 2,425 determinate valid hours reports are positive; zero genuine zero-hour reports. Because women in job-attached absence are also routed to the hours block, positive reported hours cannot be used as a current-work indicator.

C. Event-study specification

$$Y_{it} = \sum_{k \neq -2} \beta_k \mathbf{1}\{\text{event time}_{it} = k\} + \gamma_t + \varepsilon_{it}$$

- Y_{it} is current work or employment attachment in the main analysis; the appendix hours specification is interpreted only after restricting to women who report current work.
- $t = -2$ is the presentation reference period.
- γ_t are calendar-year fixed effects.
- Descriptive pooled OLS; the estimand is sample-average event-time trajectories in this analytic panel, not a causal child-penalty parameter.
- Main-slide inference: focal-woman-episode clustered SEs (no episode exclusions); robust and respondent-household clustered SEs are nearly identical.

D. Estimates with $t = -1$ as reference

Outcome (vs $t = -1$)	$t = 0$	$t = +1$	$t = +3$	$t = +5$
Current work	-0.508	-0.283	-0.163	-0.104
Employment attachment	-0.218	-0.204	-0.133	-0.133

With $t = -1$ as reference the birth-report-wave drop is smaller because the decline is already visible at $t = -1$. I therefore use $t = -2$ as the presentation reference and treat the timing profiles descriptively: the survey is annual and the birth-report wave is not an exact delivery date.

E. Inference comparison at $t = 0$ (reference $t = -2$)

Outcome	Coefficient	Robust SE	Clustered SE
Current work	-0.686	0.026	0.026
Employment attachment	-0.371	0.030	0.026

Respondent-household clustered and focal-woman-episode clustered SEs coincide in this sample (627 episodes, no episode exclusions). Inference is essentially unchanged across all three methods.

F. Weekly hours: questionnaire routing matters

The hours question is asked both to women who worked in the reference month and to women who were temporarily absent while retaining a job. For the latter, reported hours describe the job's usual hours rather than hours actually worked that month.

Coefficient vs $t = -2$	$t = -1$	$t = 0$	$t = +1$	$t = +3$	$t = +5$
All valid hours reports	-0.52	$+1.05$	-5.17	-5.84	-9.30
Currently working only	-1.01	-8.08	-7.69	-6.27	-9.35

Calendar-year fixed effects; episode-clustered SEs. At $t = 0$, 180 of 241 valid hours reporters are in job-attached absence. They report 38.8 usual hours on average, versus 27.2 hours among the 61 women who report current work. The all-valid series is therefore not a measure of realised labour supply at childbirth.

G. What do the $t = 0$ hours reports contain?

Reference-month status	Valid hours reports	Mean reported hours
Currently working	61	27.2
Job-attached absence	180	38.8

Women in job-attached absence are **not currently working**, but the questionnaire still routes them through the job-hours block. Their reported hours therefore describe the job they remain attached to rather than hours actually worked in the reference month.

Current work is therefore measured from the direct reference-month work-status item. Weekly hours are interpreted only among women who report that they are currently working.

H. Childbirth-margin risk models: full detail

Missing-indicator logits:
$\text{loss}_{t0} \sim$ nonregular + small firm + missing-contract + missing-firm; predictors at $t = -1$.

	Loss of current work	Loss of attachment
Risk set N (events)	334 (282)	357 (118)
Raw risk: regular	87.4% (153/175)	14.7% (28/191)
Raw risk: non-regular	86.9% (106/122)	58.6% (75/128)
Non-regular OR [95% CI]	1.08 [0.53, 2.21]	8.55 [4.90, 14.92]
Small-firm OR [95% CI]	0.40 [0.20, 0.78]	1.50 [0.86, 2.62]
Predicted probability: regular	86.5%	15.3%
Predicted probability: non-regular	87.3%	58.6%
AME of non-regular status	+0.8 pp	+43.1 pp

Stopping current work is common in both groups. Contract type is strongly associated with whether the employment relationship survives to $t = 0$.

I. What this design can and cannot identify

- It **can** show the timing of work interruption and attachment loss around the birth-report wave, and where attachment loss is concentrated across pre-birth contract positions.
- It **can** separate temporary, job-attached absence from loss of the employment relationship.
- It **cannot** make contract type exogenous: selection into regular employment can reflect anticipated continuity.
- It **cannot** estimate the causal effect of a specific leave reform, and event time is survey-wave time rather than time from delivery.
- "Leave or exit" is conceptual shorthand: the data identify job-attached temporary absence; childcare leave is one identified reason, and annual maternity leave is not separately identified.

J. Sample characteristics

Characteristic	Mean / share	SD / valid N
Mother age around focal birth	31.90 years	SD 4.32; $N = 630$
Partner age	33.78 years	SD 5.71; $N = 631$
Partner − mother age gap	1.89 years	SD 3.96; $N = 630$
Mother: university or higher	24.3%	valid $N = 255$
Partner: university or higher	48.4%	valid $N = 399$

“Partner” is the spouse associated with the focal birth episode; biological paternity is not directly identified.

Education uses the latest valid completed attainment observed no later than the birth-report wave. Coverage is incomplete, especially for mothers, so education shares are descriptive among observed cases.

K. Education measurement: why coverage is incomplete

Education *is* asked in the survey; the issue is measurement timing, completion status, and longitudinal coverage. Direct education questions distinguish **currently attending**, **graduated**, and **dropped out** - so a reported school level cannot automatically be read as completed attainment.

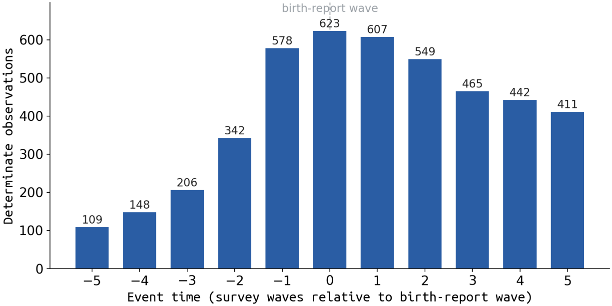
Primary / birth-wave measure

+ temporally appropriate (observed no later than the birth-report wave);
+ requires confirmed completed attainment;
− incomplete coverage: mothers $N = 255$, partners $N = 399$.

Full-history measure

+ much better coverage;
− may include education completed after childbirth;
− spouse-history identity can be ambiguous when spouse signatures change.

L. Event-time support in the unbalanced panel



Support is thinner in the tails: far leads require earlier panel entry, while far lags require continued observation. Event-study profiles are therefore descriptive sample averages, with greater composition sensitivity at longer horizons.The $t = 0$ contract comparison is within-wave; the risk models compare the same women from $t = -1$ to $t = 0$. Support is identical for current work and employment attachment.

M. Balanced short-window check ($t = -2$ to $+1$)

Same descriptive event study, restricted to the 328 women with determinate outcomes at all four waves $t = -2, -1, 0, +1$ (1,312 person-waves; calendar-year FE, $t = -2$ reference, episode-clustered SEs).

Outcome	Sample	$t = -1$	$t = 0$	$t = +1$
Current work	Unbalanced (main)	−0.177	−0.686	−0.461
	Balanced $t = -2..+1$	−0.136	−0.705	−0.422
Employment attachment	Unbalanced (main)	−0.153	−0.371	−0.357
	Balanced $t = -2..+1$	−0.098	−0.329	−0.308

The near-birth pattern holds when exactly the same women contribute to all four waves: the $t = 0$ break remains (current work slightly larger), and every coefficient keeps its sign and significance. The $t = -1$ dip is somewhat smaller, consistent with part of the sample-average pre-birth decline reflecting composition.

N. What is the attached-but-not-working state?

Determinate current-work status at $t = 0$	623
Employment-attached but not currently working	205
Explicit reported childcare leave	194
Named non-childcare reason	8
No recorded usable reason	3

Share of attached non-workers reporting childcare leave: **194/205 = 94.6%** of all cases; among rows with a recorded reason, **194/202 = 96.0%** report childcare leave. Of the 8 named non-childcare reasons, 7 are "other" and 1 is own health.

By fixed pre-birth contract group among attached non-workers: Regular **134/143 = 93.7%**; Non-regular **35/36 = 97.2%**. All 8 named non-childcare reasons occur in the regular group.

The reason item is asked only of respondents classified as temporarily absent from work while retaining a job. "For childcare leave" is a named response category. Annual maternity leave is not separately identified.

O. Why are women concentrated in non-regular employment? I

Several forces push in the same direction:

- **Regular employment is difficult to combine with care.** Long hours, overtime, transfers, and expectations of availability have historically characterised the regular-employment track (JILPT, 2019).
- **Non-regular work can offer more control over working time.** Women place particular value on flexibility in working hours when combining employment with family responsibilities (JILPT, 2019).
- **Marriage and childbirth reinforce the sorting.** Much of women's subsequent return to employment occurs through non-regular jobs, even among women who previously held regular positions (OECD, 2024).
- **Tax and social-insurance rules have historically reinforced this pattern.** Incentives facing married secondary earners encourage earnings below contribution thresholds and can encourage non-regular employment (OECD, 2024).

O. Why are women concentrated in non-regular employment? II

- **Employer demand matters too.** Firms use non-regular employment to reduce labour costs and increase workforce flexibility (OECD, 2024).

Takeaway: women facing care constraints may sort - or be pushed - into the part of the labour market that is easier to combine with family responsibilities, but that also offers weaker employment continuity.

P. What do women themselves say?

Among women working outside regular employment in a 2026 Japanese survey, the most common reasons for doing so were:

29.9%	25.2%	14.0%
"I did not think I could work in the way regular employees do."	Childcare or eldercare responsibilities.	Wanted regular employment but could not obtain it.

Interpretation

Flexibility is part of the story, but "preference" is too simple.

The concentration of women in non-regular employment reflects both care constraints and limits on access to regular jobs.

Source: JILPT, 2026.